

# Does the order of questions matter?

Two kinds of student, two ways of choosing what to ask next

28 September 2026

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## 1. What this part is about

A tutor has to decide what to ask next. There are many clever ways to decide, but before any of them is worth building we should check something simpler: *can the order possibly matter at all?*

It turns out that depends entirely on what kind of student you imagine. This document sets up two kinds of student and two ways of choosing a question, crosses them, and shows that the order is nearly irrelevant for one student and clearly important for the other.

That is a useful thing to know early. If we had only tested against the first kind of student, every scheduler we ever build would have looked the same as every other, and we would have concluded, wrongly, that scheduling does not matter.

## 2. The curriculum is a graph

The concepts are not a flat list. Some have to come before others. Our practice curriculum has eight concepts in four layers:

| layer  | concepts                                     |
|--------|----------------------------------------------|
| top    | A1, A2 (nothing comes before these)          |
| second | B1 needs A1; B2 needs A1 and A2; B3 needs A2 |
| third  | C1 needs B1 and B2; C2 needs B2 and B3       |
| last   | D1 needs C1 and C2                           |

It is shaped deliberately. B2 sits under both C1 and C2, so one weak concept blocks two later ones, and you can see it happen. And only two concepts are open at the start, so the set of things worth asking stays small enough to read.

## 3. Two kinds of student

Both students are the same underneath. Each concept is either known or not, and each time you ask about something they do not know, there is a chance they pick it up. The two differ in one respect only.

**Student S1 ignores the graph.** Asking about the last concept on day one works exactly as well as asking about it at the end. The prerequisites are drawn on the page but they do not affect this student.

**Student S2 respects the graph.** Until the prerequisites are genuinely known, this student learns the concept far more slowly: at a quarter of the usual rate. Asking too early is not useless, but it is close to wasted.

S2 is the smallest change that makes the order of questions mean something. Nothing else about the student changes.

## 4. Two ways of choosing a question

**Q1** picks at random from everything the tutor does not yet believe is mastered. It will happily ask about the last concept first.

**Q2** picks at random from the concepts whose prerequisites the tutor *believes* are already mastered. It refuses to jump ahead.

The word “believes” is doing real work there. The tutor cannot see what the student actually knows, only its own running guess. So Q2 is not following the true state of the student; it is following an estimate that lags behind.

## 5. What we expected, written down first

Before running anything, the prediction was:

|    | Q1   | Q2   |
|----|------|------|
| S1 | fine | fine |
| S2 | bad  | fine |

The reasoning: against S1 a question is worth the same whenever you ask it, so nothing the scheduler does can help much. Against S2 a question asked too early is mostly wasted, so a scheduler that avoids asking early should win.

Writing the prediction down first matters. If the scheduler had turned out to matter against S1 too, or not to matter against S2, the setup would not be able to tell schedulers apart and we would have a bug rather than a finding.

## 6. What actually happened

Four hundred simulated students, forty questions each. The same student, by seed, appears in all four boxes, so these are fair comparisons rather than four separate groups. Scored on concepts the student *truly* knows at the end, which is the student’s own state and owes nothing to what the tutor believes.

|                       | Q1   | Q2   | Q2 – Q1 | 95% CI |
|-----------------------|------|------|---------|--------|
| S1                    | 6.15 | 6.00 | –0.15   | ±0.14  |
| S2                    | 4.39 | 5.86 | +1.47   | ±0.18  |
| interaction (S2 – S1) |      |      | +1.62   | ±0.16  |

The prediction holds. Q2 is worth about one and a half concepts to S2 and is not worth anything useful to S1.

The bottom line is the real result. It says the *effect of the scheduler depends on which student you assume*. That is the claim, not either row on its own, and it is why testing a scheduler against only one kind of student can be so misleading.

One technical note. Because the same student appears in both columns, the two differences move together, and we use that:  $r = +0.49$ , and the interval would be  $\pm 0.23$  rather than  $\pm 0.16$  if the two were treated as independent.

## 7. The same result, in questions instead of concepts

Counting concepts at a fixed budget is a bit abstract. A more natural question is: how many questions before this student knows everything?

|    | Q1 | Q2 | mean diff       | median diff | favours              |
|----|----|----|-----------------|-------------|----------------------|
| S1 | 54 | 54 | $+0.3 \pm 2.3$  | +0.0        | unresolved at this n |
| S2 | 78 | 55 | $-22.4 \pm 4.3$ | -17.5       | Q2                   |

S2 gets there about twenty-three questions sooner under Q2, which is roughly a third less work. For S1 the two are level.

Every student in every box reaches all eight concepts inside the three hundred question budget, so nothing is cut off at the end and these are honest medians.

Two cautions. Every box asks the same fixed number of questions, so what differs is how quickly the student actually learns, not how much teaching was spent. And the mean and the median are not the same number here, so both are given rather than one standing in for the other.

## 8. Is it the rule, or is it the lag?

Q1 and Q2 differ in two ways at once, and it is worth separating them. Q2 *restricts* what may be asked, and it decides readiness from a belief that runs behind the truth. Either could be responsible.

So we ran a matched pair where both sides gate on the student's true state and only the restriction differs. No real tutor could do this; it is a measurement device.

|    | unrestricted | restricted | effect             |
|----|--------------|------------|--------------------|
| S1 | 7.480        | 7.480      | $+0.000 \pm 0.000$ |
| S2 | 5.513        | 7.480      | $+1.968 \pm 0.167$ |

For S1 the restriction is *exactly* neutral. Not close to zero, identical, for all four hundred students. That settles it: the small negative figure for S1 in the earlier table is not the cost of restricting what gets asked. It is the cost of deciding readiness from a belief that lags. For S2 the restriction itself is worth about two concepts.

## 9. Does it hold if we run it again?

A single run gives an interval, which tells you how well that one sample pins the number down. It does not tell you whether the same answer would come back from a fresh sample. For a small effect those are different questions, so we reran on six separate groups.

S1 averages  $+0.37 \pm 0.57$  questions of cost to Q2 over 6 independent blocks, ranging from  $-0.93$  to  $+1.08$ , but the blocks do not agree in sign, so the direction is still unsettled; S2 averages  $-23.08 \pm 1.36$  questions sooner under Q2 over 6 independent blocks, ranging from  $-25.61$  to  $-20.66$ , and every block agrees in sign, so the direction is settled.

So the S2 result is settled and the S1 result is not. Six groups put S1 slightly on the costly side of zero, but they disagree among themselves.

This is worth being blunt about, because an earlier version of this work got it wrong. We read S1 off a single group, saw a negative number, and wrote that Q2 costs S1 nothing. The interval covered zero, so that was asserting a direction the data never supported, and it happens to point the opposite

way from the estimate we now have. The honest statement is that the S1 effect is too small for us to give it a sign.

## 10. What we are not claiming

The effect sizes depend on the budget. At forty questions the scheduler gap is large; run long enough for everything to be learned anyway and every box saturates and no scheduler can differ from any other. The short budget is a deliberate choice, not an accident.

All of this is on one small eight concept curriculum, and nothing has been rerun on the larger one.

And S2 is a made up student. It is the simplest thing that makes prerequisites bite, chosen for that reason, not because we have evidence that real learners behave this way.

## 11. Running it again

`python3 make_sanity.py` reruns every study and writes the result files. `python3 make_tables.py` turns those into every table in this document, so no number here is typed by hand. `python3 make_tables.py --check` fails if the document and the results ever drift apart. `streamlit run app.py` steps through a single run one question at a time, with the graph on screen.